# ORBIT V5

# OrbitMod

Four AI Coworkers — A to Z Master Blueprint

Avis • Scout • Quill • Relay

**Core product loop**
Discover → Verify → Prove → Match → Collaborate → Build Evidence

A complete product, UX, architecture, workflow, data, security, business and implementation reference for OrbitMod V5.

Prepared as the master blueprint for the OrbitMod V5 implementation and presentation.

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# 01. Product Definition & North Star

**OrbitMod** is a Reddit-inspired professional community integrated into Orbit. It is designed to help people discover opportunities, verify who and what they are dealing with, demonstrate practical ability, find compatible collaborators, communicate safely, and turn completed work into durable professional evidence.

The product should not become a generic LinkedIn clone, a full Reddit clone, a complete ATS, or a collection of disconnected AI chatbots. The four Dots are the intelligence layer operating over one shared workspace.

> **North Star**
> Move professional networking from **claims and profiles** toward **verified opportunities, demonstrated skills and observable work**.

## Core loop

```
SPACE / OPPORTUNITY
↓
AVIS
verify trust
↓
SCOUT
match people
↓
QUILL
prove ability
↓
RELAY
communicate & collaborate
↓
PROJECT / WORK
↓
EVIDENCE PASSPORT
```

- Professionals get better evidence of what they can actually do.
- Recruiters get better signals than resumes alone.
- Teams can form around real projects instead of vague networking.
- Communities get a structured way to add useful context without turning the platform into an accusation board.
- Orbit gets a practical AI coworker layer that is tied to real objects, actions and outcomes.

# 02. The Four-Dot System

**Avis**
**Trust & Verification**

**Core question:** Can I trust this person, organisation or opportunity?

✓ Identity and organisation verification
✓ Recruiter affiliation checks
✓ Opportunity verification
✓ Evidence-backed status
✓ Disputes and appeals

**Scout**
**Discovery, Matching & Teams**

**Core question:** What opportunity or person should I connect with?

✓ Opportunity discovery
✓ Skill / interest / availability matching
✓ Project and team formation
✓ Match explanations
✓ Collaboration recommendations

**Quill**
**Skills, Assessments & Proof**

**Core question:** Can this person actually prove what they can do?

✓ Practical assessments
✓ Real-world tasks
✓ Submissions and artifacts
✓ Rubrics and reviews
✓ Evidence Passport

**Relay**
**Community, Communication & Collaboration**

**Core question:** How do people communicate and collaborate safely?

✓ Spaces, posts and comments
✓ Professional messaging
✓ Spam / scam controls
✓ Community context
✓ Team communication

**Important architecture rule**
There are **exactly four primary Dots**. Shared systems such as Evidence Passport, Opportunities and Projects are product capabilities, not extra Dots.

Each Dot should be context-aware. A Dot should not behave like a blank chat box. It should understand the current object and offer relevant actions.

# 03. Avis — Trust & Verification

**Core question**
Can I trust this person, organisation or opportunity?

Avis is the trust layer. Its job is to collect and explain evidence, not to pretend that an AI model can prove authenticity with certainty. Verification should be a combination of explicit checks, provenance where available, account signals, organisation evidence and human review when required.

## Primary objects

| Object | Purpose |
|---|---|
| Identity | User verification state and supporting evidence. |
| Organisation | Verified company / institution record. |
| Recruiter affiliation | Relationship between a person and an organisation. |
| Opportunity | Job, internship, project or event listing. |
| Verification | Checks, evidence, state, reviewer and timestamps. |
| Appeal | Challenge to a verification or moderation decision. |

## Capabilities

- Identity verification and verification state management
- Organisation domain / ownership checks where supported
- Recruiter affiliation verification
- Opportunity source consistency checks
- Evidence attachment and provenance metadata
- Risk / suspicious activity signals
- Human-review queue for ambiguous cases
- Dispute and appeal workflow
- Audit trail for verification actions
- Clear explanation of why a status exists

## Primary workflow

```
Opportunity created
↓
Collect organisation + recruiter evidence
↓
Run deterministic checks
↓
Run risk signals where appropriate
↓
If ambiguous → human review
↓
VERIFIED / LIMITED EVIDENCE / NEEDS REVIEW / SUSPICIOUS / UNVERIFIED
↓
Expose explanation + evidence
```

## Core screens

- Verification detail screen

- Organisation verification profile
- Recruiter verification panel
- Opportunity trust panel
- Review queue
- Appeal / dispute screen
- Audit history

## Key inputs

- Organisation domain
- Recruiter email / affiliation
- Public organisation information
- Opportunity source
- Uploaded verification evidence
- System risk signals

## Key outputs

- Verification state
- Evidence list
- Explanation
- Reviewer status
- Last checked timestamp
- Appeal state

## Important edge cases

- A legitimate organisation has no supported domain evidence
- Recruiter affiliation changes after verification
- Evidence expires
- False positive suspicious signal
- User appeals a decision
- A verification provider is unavailable
- Sensitive evidence must remain private

## Success metrics

- Verification completion rate
- Median verification time
- Appeal rate
- False-positive rate after human review
- Percentage of opportunities with clear evidence
- Report-to-resolution time

**Future extensions**
Later: pluggable third-party identity and media-provenance providers, stronger organisation verification, device / account risk signals and richer provenance analysis. Do not build a fake deepfake detector in V5.

# 04. Scout — Discovery, Matching & Teams

**Core question**
What opportunity or person should I connect with?

Scout converts the platform's structured information into explainable recommendations. It should match people to opportunities and projects using observable signals such as skills, interests, availability and evidence. It must not claim perfect personality or compatibility prediction.

## Primary objects

| Object | Purpose |
|---|---|
| Opportunity | A job, internship, project, hackathon or collaboration. |
| Match | A ranked recommendation with reasons. |
| Team | A group formed around a project or goal. |
| Team member | A user with relevant skills / availability. |
| Project | A shared work context created after matching. |

## Capabilities

- Opportunity discovery and ranking
- Skill matching
- Interest matching
- Availability matching
- Evidence-aware matching
- Person ↔ opportunity recommendations
- Person ↔ project recommendations
- Complementary team formation
- Match explanations
- Invite and accept / decline flow
- Basic trial collaboration support

## Primary workflow

```
User / project requirements
↓
Normalize required skills + constraints
↓
Retrieve candidate opportunities / people
↓
Score observable signals
↓
Generate explanation
↓
Rank results
↓
User accepts / rejects / saves
↓
Optional team formation
```

## Core screens

- Scout home / recommendations
- Opportunity search
- Match explanation panel
- People discovery
- Team builder
- Team invitation flow
- Project creation

## Key inputs

- Required skills
- User skills / evidence
- Availability
- Interests
- Project constraints
- Opportunity type
- Location / remote preference

## Key outputs

- Ranked results
- Match reasons
- Compatibility dimensions
- Suggested teammates
- Team composition suggestions

## Important edge cases

- Sparse profile
- Conflicting availability
- No strong match
- Skill claim without evidence
- Candidate opts out
- Ranking bias
- Overfitting to historical activity

## Success metrics

- Recommendation click-through
- Save / apply rate
- Match acceptance rate
- Team formation completion
- Time to first relevant opportunity
- Post-match project completion

**Future extensions**
Later: richer semantic matching, organisation-specific hiring criteria, team balance optimization and integrations with external opportunity systems.

# 05. Quill — Skills, Assessments & Evidence

**Core question**
Can this person actually prove what they can do?

Quill is the proof-of-work engine. It replaces vague skill claims with practical demonstrations. The assessment system should prioritize realistic tasks, transparent rubrics, observable artifacts and reviewer feedback.

## Primary objects

| Object | Purpose |
|---|---|
| Assessment | A practical challenge with requirements and rubric. |
| Submission | Candidate's completed response / artifact. |
| Artifact | Code, design, document, presentation or other work. |
| Rubric | Transparent evaluation criteria. |
| Evidence | Persistent record of demonstrated ability. |
| Review | Human or AI-assisted evaluation. |

## Capabilities

- Practical coding tasks
- Design challenges
- Business / marketing cases
- Project simulations
- Custom organisation assessments
- Time limits and instructions
- Artifact collection
- Rubric-based evaluation
- AI-assisted feedback
- Human review
- Evidence creation
- Portfolio presentation

## Primary workflow

```
Opportunity requires proof
↓
Quill creates / selects assessment
↓
Candidate receives task + rubric
↓
Candidate submits work
↓
Artifact is stored securely
↓
AI assists evaluation
↓
Human review when required
↓
Result + evidence record
↓
Evidence Passport updated
```

## Core screens

- Assessment library
- Assessment builder
- Candidate task screen
- Submission workspace
- Reviewer screen
- Rubric editor
- Evidence detail
- Evidence Passport

## Key inputs

- Task requirements
- Candidate identity
- Submission files / links
- Time constraints
- Rubric
- Reviewer input

## Key outputs

- Submission
- Score / qualitative result
- Artifact
- Reviewer feedback
- Evidence record
- Skill association

## Important edge cases

- Plagiarized or copied work
- Incomplete submission
- Broken deployment link
- AI-generated submission
- Reviewer disagreement
- Expired assessment

- Sensitive employer assessment leakage

## Success metrics

- Assessment completion
- Submission rate
- Reviewer agreement
- Evidence creation rate
- Time-to-review
- Candidate satisfaction
- Correlation between assessment outcome and project outcome

**Future extensions**
Later: proctoring integrations, richer sandbox execution, organization-specific assessment libraries, skills graphs and portable evidence credentials.

# 06. Relay — Community, Communication & Collaboration

**Core question**
How do people communicate and collaborate safely?

Relay owns the social and communication layer. It combines Reddit-style Spaces, professional messaging, moderation controls, community context and project communication. Safety is a capability inside Relay, not a separate Dot.

## Primary objects

| Object | Purpose |
|---|---|
| Space | A community organized around a topic or goal. |
| Post | A community discussion or opportunity post. |
| Comment | Response to a post. |
| Message | Private professional communication. |
| Report | User-submitted safety or policy signal. |
| Community note | Evidence-backed context on a public claim. |

## Capabilities

- Spaces and memberships
- Posts, comments and voting
- Opportunity discussions
- Direct messages
- Communication permissions
- Spam / scam classification
- Suspicious-link warnings
- Message quarantine
- Reports and appeals
- Evidence-backed community context
- Organisation response
- Team project communication

## Primary workflow

```
Community / project event
↓
Relay receives content
↓
Apply permissions + safety checks
↓
DELIVER / FILTER / QUARANTINE
↓
User can report / appeal
↓
Moderation workflow
↓
Context retained in audit trail
```

## Core screens

- Space home
- Post composer
- Post detail
- Comment thread
- Messages
- Message request inbox
- Moderation queue
- Community context panel
- Organisation response panel
- Project chat

## Key inputs

- Message text
- Sender relationship
- Space membership
- Link metadata
- User reports
- Community evidence

## Key outputs

- Delivered message
- Filtered message
- Quarantined message
- Report
- Community context
- Organisation response
- Appeal state

## Important edge cases

- False positive spam detection
- Coordinated brigading
- Harassment
- Unsupported accusation
- Malicious link
- Mass messaging

- Organisation disputes note
- Moderator overload

## Success metrics

- Healthy message delivery rate
- Spam catch rate
- False-positive rate
- Report resolution time
- Community contribution rate
- Appeal resolution time

**Future extensions**
Later: richer moderation models, community contributor reputation, organization moderation controls and cross-community safety signals.

# 07. Shared OrbitMod Systems

The four Dots operate on common product objects. This is what makes OrbitMod a coherent system rather than four separate assistants.

| System | Purpose | Primary Dots |
|---|---|---|
| Spaces | Community discovery and discussion | Relay + Scout |
| Opportunities | Jobs, internships, projects, hackathons | Avis + Scout + Quill + Relay |
| Evidence Passport | Persistent proof of work | Quill + Scout |
| Teams | People assembled around a goal | Scout + Relay |
| Projects | Lightweight collaboration workspace | Scout + Relay + Quill |
| Profiles | Identity, skills, evidence and activity | Avis + Scout + Quill |
| Messages | Professional communication | Relay |
| Community context | Evidence-backed public context | Relay + Avis |
| Notifications | Events requiring attention | All four |
| Audit log | Traceability of sensitive actions | All four |

## Evidence Passport

The Evidence Passport is the central professional proof layer. It should show what a person has actually completed, not simply what they claim.

```
PROFILE
■■ Identity / verification
■■ Skills
■■ Evidence
■ ■■ Practical assessments
■ ■■ Projects
■ ■■ Artifacts
■ ■■ Reviewer feedback
■■ Teams
■■ References (later / optional)
```

## Project Workspace

Keep projects intentionally lightweight in V5. They need enough structure to move from a match to actual work: members, tasks, milestones, discussion, files, deliverables and evidence. Do not turn OrbitMod into a full Jira, Slack or Notion replacement.

# 08. End-to-End User Journeys

## A. Candidate finds an internship

```
Space
→ Opportunity
→ Avis verification
→ Scout match explanation
→ Apply
→ Quill practical assessment
→ Submission
→ Evidence created
→ Relay communication
```

## B. Founder forms a project team

```
Founder creates project requirements
→ Scout identifies complementary skills
→ Invite candidates
→ Team accepts
→ Project workspace created
→ Relay collaboration
→ Quill evidence generated from deliverables
```

## C. User checks a suspicious opportunity

```
Open opportunity
→ Ask Avis
→ Review organisation + recruiter evidence
→ See status and explanation
→ Report / appeal if necessary
→ Relay surfaces relevant community context
```

## D. Recruiter validates a candidate

```
Open candidate
→ Review Avis identity / organisation context
→ Review Quill evidence
→ Compare practical assessment
→ Scout suggests fit
→ Relay starts professional conversation
```

## E. Community flags a misleading claim

```
Post published
→ Relay receives community context
→ User attaches evidence
→ Note enters review
→ Organisation can respond
→ Final status shown with provenance and audit trail
```

**Golden demo story**

For a judge demo, use one continuous narrative: **discover an opportunity** → **verify it** → **match a candidate** → **complete a practical test** → **form a team** → **collaborate** → **create evidence**.

# 09. UX / UI Architecture

OrbitMod should feel like a natural evolution of Orbit. The interface should be information-dense and professional without becoming visually noisy.

## Primary navigation

```
ORBIT
■■■ Home
■■■ OrbitMod
■
■■■ SPACES
■ ■■■ All Spaces
■ ■■■ My Spaces
■ ■■■ Joined
■
■■■ DOTS
■ ■■■ Avis
■ ■■■ Scout
■ ■■■ Quill
■ ■■■ Relay
■
■■■ WORKSPACE
■■■ Opportunities
■■■ Projects
■■■ Evidence
```

## Core screens

| Screen | What it should answer |
|---|---|
| OrbitMod Home | What should I do or explore next? |
| Space | What is happening in this community? |
| Opportunity | Is this legitimate and is it relevant to me? |
| Avis | What evidence supports the trust state? |
| Scout | Why is this recommended? |
| Quill | What must I demonstrate and how will it be judged? |
| Relay | Who can contact me and what needs attention? |
| Project | What are we building and what remains? |
| Evidence Passport | What can I prove I have done? |

## Design principles

- Use Orbit's existing design tokens and components.
- Keep Dot identities visually distinct but structurally consistent.
- Prefer evidence, labels and explanations over decorative AI effects.
- Do not use fake metrics, fake testimonials or fabricated activity.
- Avoid excessive glassmorphism, giant gradients, pill-shaped UI everywhere and over-animated pages.
- Empty states should explain what the user can do next.
- Important trust decisions should show reasons, not just colored badges.

# 10. Technical Architecture

The V5 architecture should extend the existing Orbit stack. Avoid a rewrite and avoid microservices unless the current application already uses them.




## Recommended technology baseline

| Layer | Recommendation | Rule |
|---|---|---|
| Frontend | Existing Orbit frontend; Next.js + TypeScript if already used | Reuse existing app |
| UI | Existing component library / Tailwind / shadcn if already present | Do not replace design system |
| Backend | Existing Orbit API / server actions | Extend existing patterns |
| Database | PostgreSQL | Use existing schema |
| ORM | Existing ORM (Prisma if already used) | Do not introduce a second ORM |
| Vector matching | pgvector when genuinely useful | Optional |
| Storage | Existing object storage / S3-compatible storage | Private by default |
| Realtime | Existing realtime system / WebSockets if needed | Only where required |
| Queue | Existing worker / Redis if already present | Use for async AI and notifications |
| Validation | Zod or existing validation library | Consistent API contracts |
| Testing | Existing test stack + Playwright for critical flows | Prioritize end-to-end path |

## Architecture rule

Use one shared AI gateway rather than four independent AI implementations. Each Dot should call domain-specific tools and workflows through the same controlled AI layer.

# 11. Data Model

This is a conceptual model. Before implementation, inspect the existing Orbit database and reuse equivalent entities rather than creating duplicates.

```
User
■■ Profile
■■ Skills
■■ Evidence
■■ Memberships
■■ Messages

Organisation
■■ Verification
■■ Recruiters
■■ Opportunities

Space
■■ Membership
■■ Posts
■■ Comments

Opportunity
■■ Organisation
■■ Verification
■■ Assessment
■■ Applications

Assessment
■■ Rubric
■■ Submission
■■ Artifact
■■ Evidence

Team
■■ Members
■■ Project
■■ Tasks
■■ Milestones
■■ Files
■■ Deliverables

CommunityNote
■■ Evidence
■■ OrganisationResponse
■■ Appeal

AuditEvent
■■ Actor
■■ Action
■■ Resource
```

## Minimum fields worth considering

| Entity | Core fields |
|---|---|
| User | id, profile, settings, createdAt |
| Organisation | id, name, domain, verificationState |
| Space | id, name, description, visibility |
| Opportunity | id, organisationId, type, title, requirements, status |
| Verification | id, resourceType, resourceId, state, evidence, reviewer, timestamps |

| Entity | Core fields |
| --- | --- |
| Assessment | id, opportunityId, instructions, rubric, duration, status |
| Submission | id, assessmentId, candidateId, artifacts, submittedAt, review |
| Evidence | id, ownerId, type, source, artifact, reviewers, visibility |
| Team | id, projectId, members, status |
| Project | id, title, requirements, tasks, milestones, deliverables |
| Message | id, senderId, recipientId / channel, body, status |
| CommunityNote | id, targetId, claim, context, evidence, status, response |
| AuditEvent | id, actorId, action, resource, timestamp |

# 12. API & Event Design

The API should follow existing Orbit conventions. Names below are conceptual examples, not mandatory routes.

| Domain | Example operations |
|---|---|
| Spaces | GET /spaces, GET /spaces/:id, POST /spaces/:id/posts |
| Opportunities | GET /opportunities, POST /opportunities, GET /opportunities/:id |
| Avis | POST /verifications, GET /verifications/:id, POST /appeals |
| Scout | POST /matches, GET /recommendations, POST /teams |
| Quill | POST /assessments, POST /submissions, POST /reviews |
| Evidence | GET /evidence, POST /evidence, PATCH /evidence/:id |
| Relay | GET /messages, POST /messages, POST /reports |
| Projects | POST /projects, GET /projects/:id, POST /projects/:id/tasks |

## Important events

```
opportunity.created
verification.requested
verification.completed
match.generated
team.created
assessment.assigned
assessment.submitted
assessment.reviewed
evidence.created
message.sent
message.quarantined
community_note.created
appeal.created
project.completed
```

## Event principles

- Events should be idempotent where possible.
- Sensitive events should be permission-aware.
- Do not put private content into logs or analytics by default.
- AI jobs should be asynchronous when they are expensive or non-interactive.
- Every user-visible trust decision should be traceable to a current state and evidence.

# 13. AI Architecture

AI is an assistant layer over structured product data. The Dots should not become unconstrained chatbots.

| Dot | AI jobs | Deterministic jobs |
|---|---|---|
| Avis | Summarise evidence, surface anomalies, explain verification context | Permission checks, state transitions, audit logs |
| Scout | Semantic matching, recommendation explanations | Filters, eligibility, availability constraints |
| Quill | Generate assessment drafts, assist rubric review, summarise feedback | Submission deadlines, artifact storage, evidence creation |
| Relay | Intent classification, spam/scam signals, content summarisation | Messaging permissions, quarantine states, reporting workflow |

## AI guardrails

- AI must not silently decide a user's identity status.
- AI must not make final hiring decisions.
- AI-generated assessment feedback should be labeled appropriately.
- AI should cite or point to the underlying evidence when making a recommendation.
- Do not store sensitive prompts or content longer than necessary.
- Use structured outputs and schema validation.
- Provide fallback behavior when the AI provider fails.
- Keep model/provider configuration behind one abstraction.

**Rule of thumb**
If the same outcome can be produced deterministically, do it deterministically. Use AI for interpretation, generation, ranking support and explanation — not for basic authorization or state management.

# 14. Security, Privacy & Abuse Prevention

OrbitMod deals with identity, employment information, private messages, assessment artifacts and organisation evidence. Security is therefore a core product requirement, not a later polish step.

| Area | Requirement |
| --- | --- |
| Authentication | Reuse Orbit authentication; support secure sessions. |
| Authorization | Resource-level access checks for every sensitive object. |
| Private evidence | Never expose by default; use signed/private access. |
| Messages | Only participants / authorized moderators can access private messages. |
| Assessments | Candidate and authorized reviewers only. |
| Verification docs | Private; never public by accident. |
| Uploads | Validate type, size, extension and storage permissions. |
| Secrets | Environment / secret manager only; never client-side. |
| Rate limiting | Protect messaging, reports, verification and AI endpoints. |
| Abuse | Reports, quarantine, blocks and appeals. |
| Audit | Record sensitive state changes without logging secrets or private bodies. |
| AI safety | Treat external content as untrusted input; prevent prompt injection from controlling privileged actions. |

## Threat scenarios

- Fake recruiter attempts to move users off-platform.
- Attacker uploads malicious file as assessment evidence.
- User attempts to access another candidate's submission.
- Coordinated group brigades a company or opportunity.
- Spam bot mass-sends opportunity links.
- Prompt injection tries to make a Dot reveal private data.
- Compromised recruiter account posts a fraudulent opportunity.

# 15. Business Model

OrbitMod should monetize the professional workflow, not the trust decision itself. Users should be able to discover opportunities and build a basic evidence record without paying to become 'more trusted'.

| Revenue stream | Customer | Value |
|---|---|---|
| Employer / recruiter subscription | Startups, companies, recruiters | Matching, assessments, candidate evidence, hiring workflow |
| Quill assessments | Employers / organisations | Practical evaluation infrastructure |
| Campus / university plans | Universities and institutions | Private Spaces, assessments, hackathons, employer connections |
| Team / project workspace | Startups and teams | Advanced collaboration features |
| Sponsored opportunities | Verified organisations | Reach, with verification independent from payment |
| Hackathon / event platform | Organisers | Challenges, submissions, team discovery |
| API / integrations (later) | ATS / HR / platforms | Verification, matching and assessment infrastructure |

**Trust rule**
Payment must never buy verification, remove legitimate reports, hide community evidence or artificially improve trust status.

## Suggested packaging

| Plan | Indicative scope |
|---|---|
| Free | Spaces, discovery, basic profile, basic evidence |
| Pro | Advanced evidence, more assessments, deeper matching, richer workspace |
| Business | Recruiter seats, custom assessments, organisation tools, analytics |
| Campus | Private community, student programs, assessments, events and employer workflows |

# 16. Metrics & Success Criteria

Focus on outcome metrics, not vanity metrics.

| Area | Metric | Why it matters |
|---|---|---|
| Discovery | Relevant opportunity click-through | Measures recommendation usefulness |
| Trust | Verified opportunity rate | Shows trust layer adoption |
| Trust | Appeal / false-positive rate | Tests whether verification is fair |
| Proof | Assessment completion rate | Measures Quill engagement |
| Proof | Evidence creation rate | Measures durable value |
| Matching | Match acceptance rate | Measures Scout usefulness |
| Teams | Team formation rate | Measures collaboration conversion |
| Projects | Project completion rate | Measures real-world outcomes |
| Communication | Report / false-positive rate | Measures Relay safety quality |
| Community | Useful context contributions | Measures community signal quality |
| Business | Paid organisation retention | Measures commercial value |

**V5 success is not 'we have many features'. V5 success is a judge or user being able to complete the core loop without confusion.**

# 17. MVP / V5 Scope

V5 should be deliberately constrained. Build one coherent system rather than shallow versions of every possible marketplace feature.

## Must-have

- OrbitMod navigation and home
- Spaces with posts / comments
- Opportunity object
- Avis verification states and explanation
- Scout basic matching and explanations
- Quill one or two practical assessment patterns
- Submission → Evidence Passport
- Relay professional messaging and basic safety controls
- Lightweight Team + Project workspace
- End-to-end demo flow
- Authentication and permissions
- Responsive UI

## Should wait

- Production-grade deepfake detection
- Complex social reputation algorithms
- Full ATS
- Full project management suite
- Advanced HR analytics
- Large-scale external integrations
- Complex community consensus algorithms
- Multi-provider identity orchestration

**Scope filter**
If a feature does not materially improve **Discover** → **Verify** → **Prove** → **Match** → **Collaborate** → **Evidence**, it does not belong in V5.

# 18. Implementation Plan

| Phase | Deliverable |
|---|---|
| Phase 0 — Inspect | Map existing Orbit architecture, routes, auth, DB, components, AI layer and existing Dot system. |
| Phase 1 — Shell | OrbitMod routes, sidebar, home, Spaces and basic navigation. |
| Phase 2 — Opportunities | Opportunity creation, browsing, detail page and status. |
| Phase 3 — Avis | Verification object, states, evidence panel, review flow and appeal. |
| Phase 4 — Scout | Recommendations, match explanations, team creation and invitations. |
| Phase 5 — Quill | Assessment, candidate submission, artifact storage, review and evidence. |
| Phase 6 — Relay | Spaces, posts, messages, permissions, filtering and reporting. |
| Phase 7 — Projects | Lightweight team workspace, tasks, milestones and deliverables. |
| Phase 8 — Integration | Connect the full flow and shared notifications. |
| Phase 9 — Polish | Responsive UX, empty/error/loading states, accessibility and visual consistency. |
| Phase 10 — QA | Automated tests, manual end-to-end test, build verification and regression testing. |

## Definition of done

- Existing Orbit functionality remains intact.
- Every new route has loading, error and empty states.
- Every sensitive resource has access control.
- Core workflow works with real persisted data.
- No fake metrics or fake testimonials are needed.
- AI failure does not break the core product.
- Mobile and desktop flows are usable.

# 19. Testing & QA

Test the product as a workflow, not only as isolated pages.

| Test layer | Examples |
|---|---|
| Unit | Matching score logic, verification state transitions, permission helpers |
| API | Create opportunity, submit assessment, create evidence, send message |
| Security | Unauthorized evidence access, private message access, file permissions |
| AI | Schema validation, fallback, prompt injection resistance |
| E2E | Space → Opportunity → Avis → Scout → Quill → Evidence → Team → Relay |
| UI | Responsive layout, keyboard navigation, empty / loading / error states |
| Regression | Existing Orbit login, home, profile, Dot functionality |

## Critical acceptance test

```
Given a signed-in user
When they open an OrbitMod Space
And open a verified opportunity
And receive a Scout recommendation
And complete a Quill assessment
And submit an artifact
Then an Evidence record is created
And the user can view it in the Evidence Passport
And Scout can use the evidence for a project match
And Relay can support collaboration
And existing Orbit features still work.
```

# 20. Roadmap

| Version | Focus |
| --- | --- |
| V5 | Core OrbitMod loop: Spaces, Opportunities, Avis, Scout, Quill, Relay, Evidence, lightweight Projects |
| V5.1 | Better matching explanations, richer assessment library, stronger moderation controls, improved analytics |
| V5.2 | Organisation workflows, campus programs, event / hackathon support |
| V6 | External integrations, ATS / developer integrations, richer evidence portability |
| Later | Advanced verification providers, media provenance, API platform, larger ecosystem |

## Roadmap principle

Only scale infrastructure and feature complexity after the core workflow shows real usage. Avoid building speculative systems because they sound impressive in a pitch.

# 21. Judge / Demo Narrative

The cleanest presentation is a single story rather than a feature list.

```
PROBLEM
People discover opportunities but cannot easily trust them,
prove skills, find the right collaborators or collaborate safely.

SOLUTION
OrbitMod integrates four AI coworkers into Orbit.

AVIS
"Can I trust this?"

SCOUT
"Who or what should I connect with?"

QUILL
"Can I prove I can do it?"

RELAY
"How do we communicate and collaborate safely?"

DEMO
Space
→ Opportunity
→ Verify
→ Match
→ Practical Test
→ Evidence
→ Team
→ Project
→ Outcome

BUSINESS
Professionals use the core network.
Organisations pay for hiring, assessments,
verification workflows and collaboration tools.

VISION
Move professional networking from claims
toward verified work and real outcomes.
```

## Suggested 7-minute demo

| Time | Action |
|---|---|
| 0:00–0:45 | Problem + one-sentence solution |
| 0:45–1:30 | OrbitMod home + Spaces |
| 1:30–2:15 | Open opportunity + Avis |
| 2:15–3:00 | Scout match explanation |
| 3:00–4:15 | Quill practical assessment + evidence |
| 4:15–5:15 | Team + Relay collaboration |
| 5:15–6:15 | Evidence Passport |
| 6:15–7:00 | Business model + differentiation + closing |

# 22. Feature Matrix & Final Checklist

| Capability | Avis | Scout | Quill | Relay |
|---|---|---|---|---|
| Identity / organisation trust | ✓ | | | |
| Opportunity verification | ✓ | ✓ | | |
| Opportunity discovery | | ✓ | | ✓ |
| Skill matching | | ✓ | ✓ | |
| Team formation | | ✓ | | ✓ |
| Practical assessment | | | ✓ | |
| Evidence Passport | | ✓ | ✓ | |
| Spaces | | | | ✓ |
| Posts / comments | | | | ✓ |
| Professional messaging | | | | ✓ |
| Spam / scam controls | | | | ✓ |
| Community context | ✓ | | | ✓ |
| Project collaboration | | ✓ | ✓ | ✓ |
| Appeals | ✓ | | | ✓ |

## Final product checklist

- OrbitMod is integrated into the existing Orbit application.
- Exactly four primary Dots exist: Avis, Scout, Quill, Relay.
- Avis is the trust layer and does not pretend AI guarantees authenticity.
- Scout provides explainable discovery and matching.
- Quill converts practical work into evidence.
- Relay owns community, communication and safety.
- Evidence Passport is shared across the four Dots.
- Opportunities connect the four Dots into one workflow.
- Teams and Projects connect matching to actual work.
- Community context is evidence-backed and supports organisation response.
- Private data is protected with resource-level permissions.
- The V5 implementation is intentionally smaller than a full LinkedIn / Reddit / ATS replacement.
- The complete end-to-end workflow works with real persisted data.
- Existing Orbit functionality remains stable.

**ONE-LINE SUMMARY**
OrbitMod is a professional community inside Orbit where **Avis builds trust, Scout finds the right connections, Quill proves real skills, and Relay enables safe collaboration — turning opportunities into verified work and evidence.**

End of master blueprint.